Setup:

First Create venv: python -m venv .venv

• Activate: Windows: .venv\Scripts\activate | macOS/Linux: source .venv/bin/activate

• Install packages: pip install -r requirements.txt

• Run: python main.py (or python app.py depending on your file name)

• Freeze: pip freeze > requirements.txt

Project Features

1. Data Entry Form

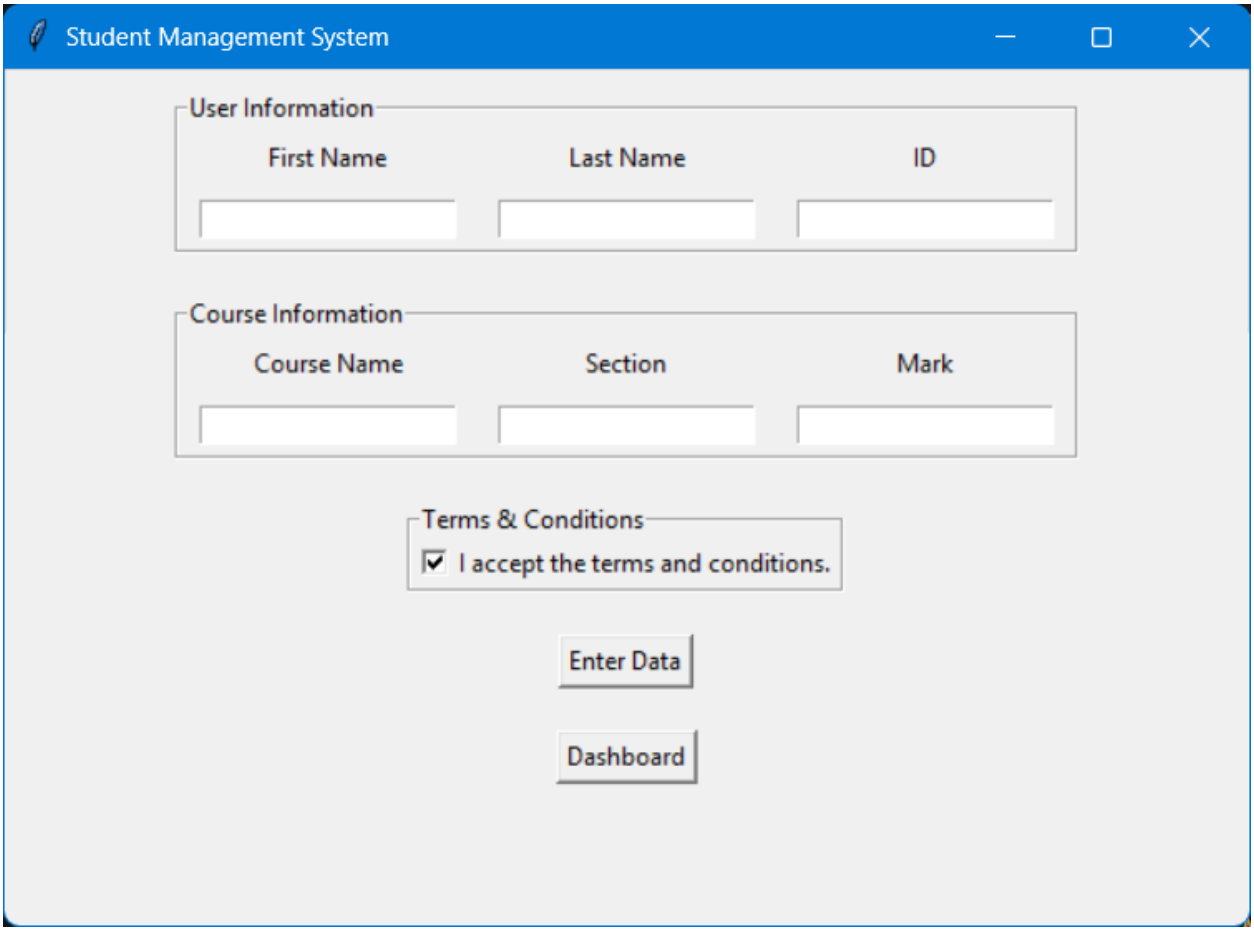
- Enter Student Information:
  - First Name
  - Last Name
  - Unique Student ID (duplicate IDs are not allowed)
- Enter Course Information:
  - Course Name
  - Section
  - Mark
- Terms & Conditions acceptance before saving.
- Data Validation:
  - All fields are required.
  - Prevent duplicate Student IDs.
- Data Persistence:
  - Saves all information to an Excel file (`data.xlsx`) using openpyxl.
- Clear Form after submission.

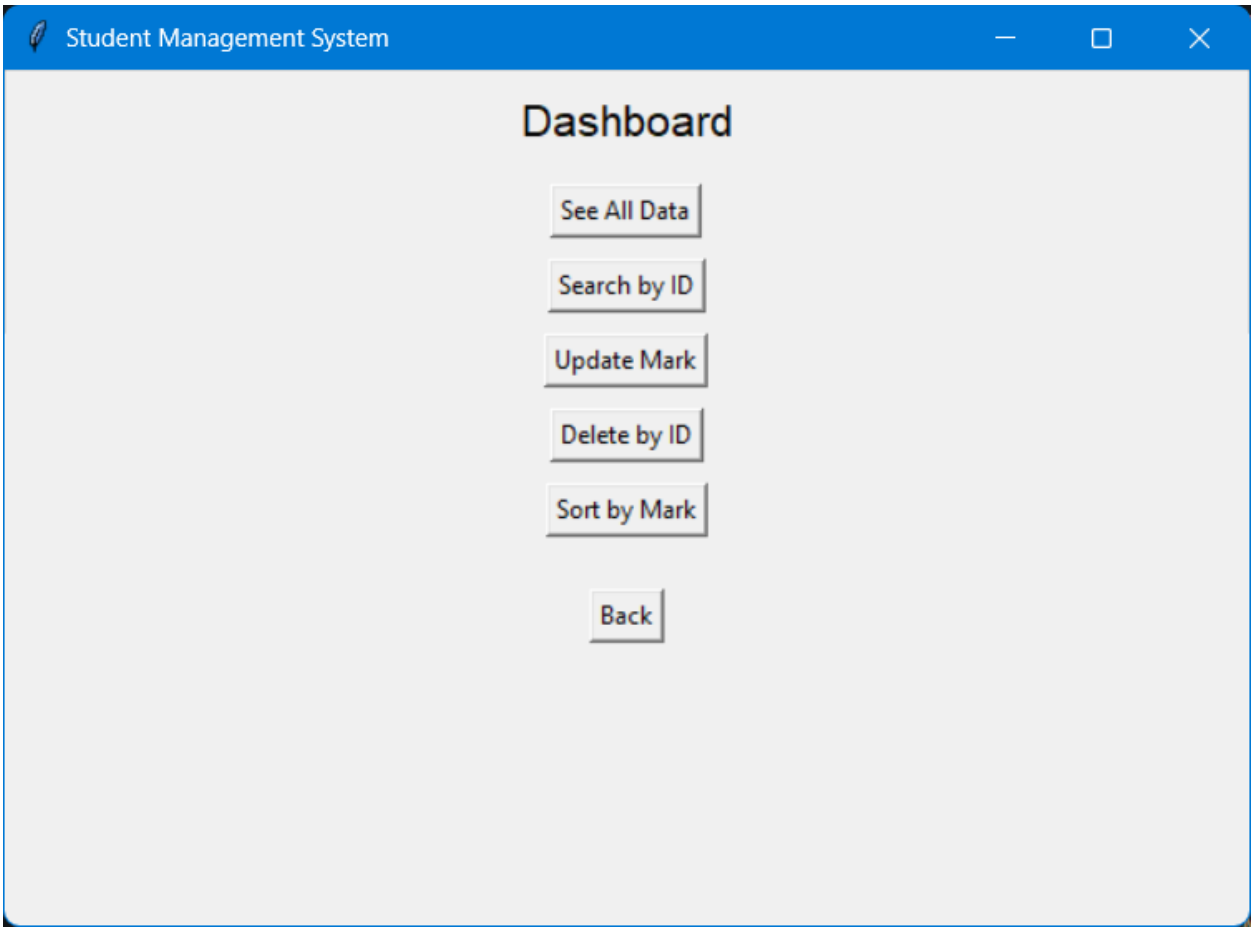
2. Dashboard

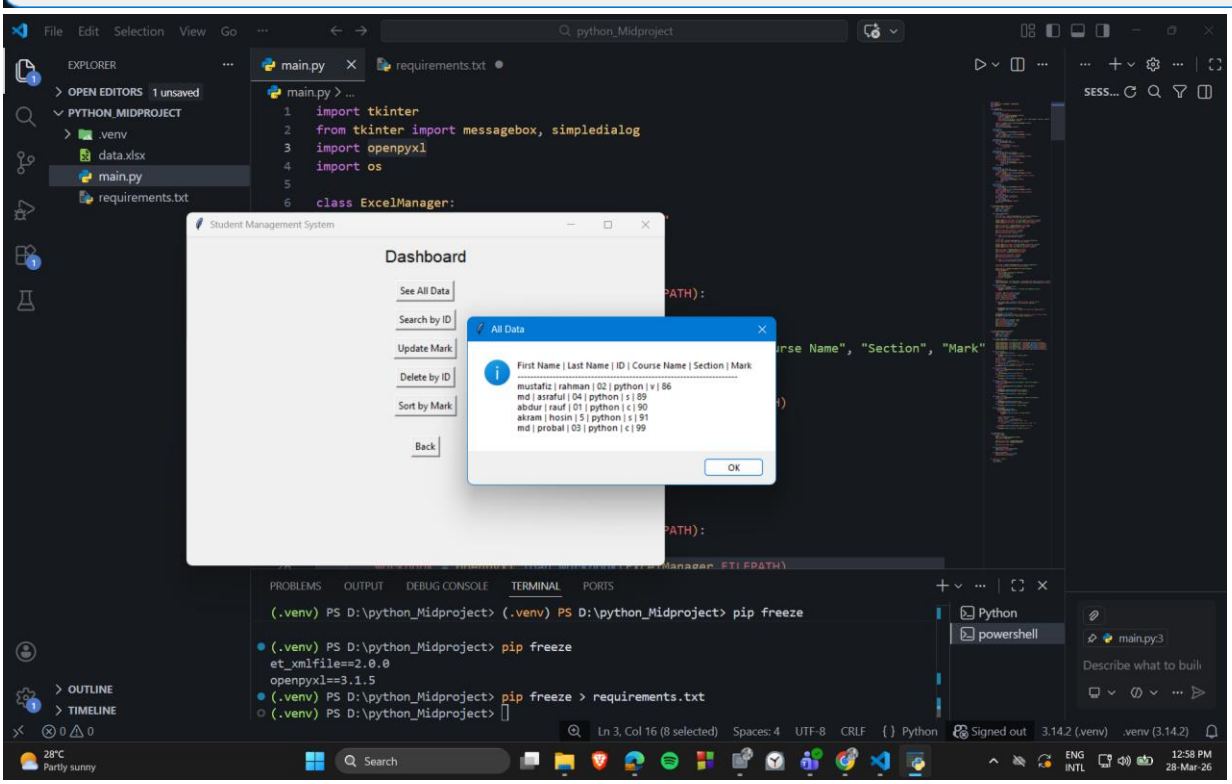
- See All Data:
  Displays all student records in a popup in a table-like format.
- Search by ID:

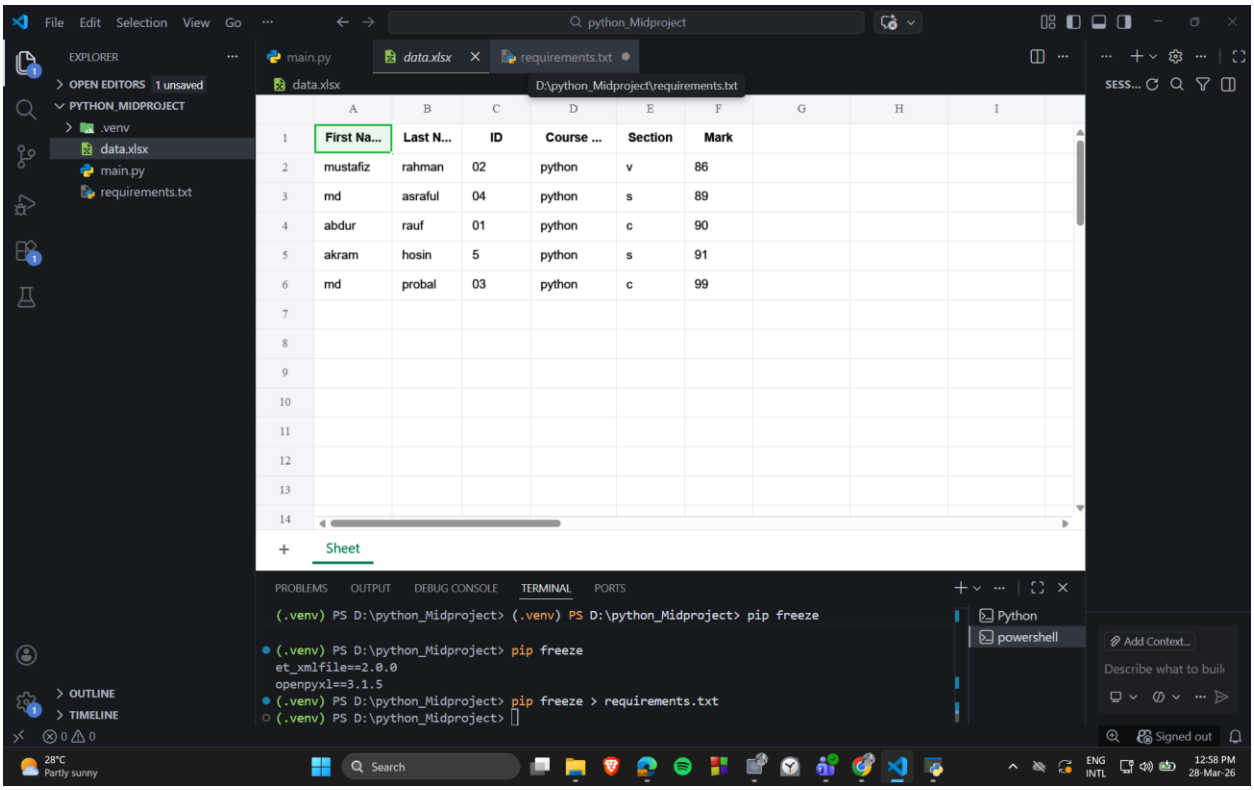
Search for a student with their unique ID and show their record in a popup.

- Update Mark:
  Update the mark of a student by ID.
- Delete by ID:
  Delete a student record with an ID.
- Sort by Mark:
  Sort all student records by marks in ascending order.
  Displays sorted records in a popup in a table-like format.
- Back Button:
  Navigate back to the data entry form.




Student Management System
Dashboard
See All Data
Search by ID
Update Mark
Delete by ID
Sort by Mark
Back


main.py
requirements.txt
import tkinter
from tkinter import messagebox, simpledialog
import openpyxl
import os
class ExcelManager:
Student Management System
Dashboard
See All Data
Search by ID
Update Mark
Delete by ID
Sort by Mark
Back
All Data
First Name | Last Name | ID | Course Name | Section | Mark
mustafiz | rahman | 02 | python | v | 86
md | asraful | 04 | python | s | 89
abdur | rauf | 01 | python | c | 90
akram | hosin | 5 | python | s | 91
md | probal | 03 | python | c | 99
OK
(.venv) PS D:\python_Midproject> pip freeze
et_xmlfile==2.0.0
openpyxl==3.1.5
(.venv) PS D:\python_Midproject> pip freeze > requirements.txt


File Edit Selection View Go
python_Midproject
EXPLORER
OPEN EDITORS 1 unsaved
PYTHON_MIDPROJECT
.venv
data.xlsx
main.py
requirements.txt
main.py
data.xlsx
requirements.txt
D:\python_Midproject\requirements.txt
First Na... Last N... ID Course ... Section Mark
mustafiz rahman 02 python v 86
md asraful 04 python s 89
abdur rauf 01 python c 90
akram hosin 5 python s 91
md probal 03 python c 99
Sheet
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(.venv) PS D:\python_Midproject> (.venv) PS D:\python_Midproject> pip freeze
(.venv) PS D:\python_Midproject> pip freeze
et_xmlfile==2.0.0
openpyxl==3.1.5
(.venv) PS D:\python_Midproject> pip freeze > requirements.txt
(.venv) PS D:\python_Midproject>
Python
powershell
OUTLINE
TIMELINE
Add Context...
Describe what to build
Signed out
28°C
Partly sunny
Search
ENG
INTL
12:58 PM
28-Mar-26